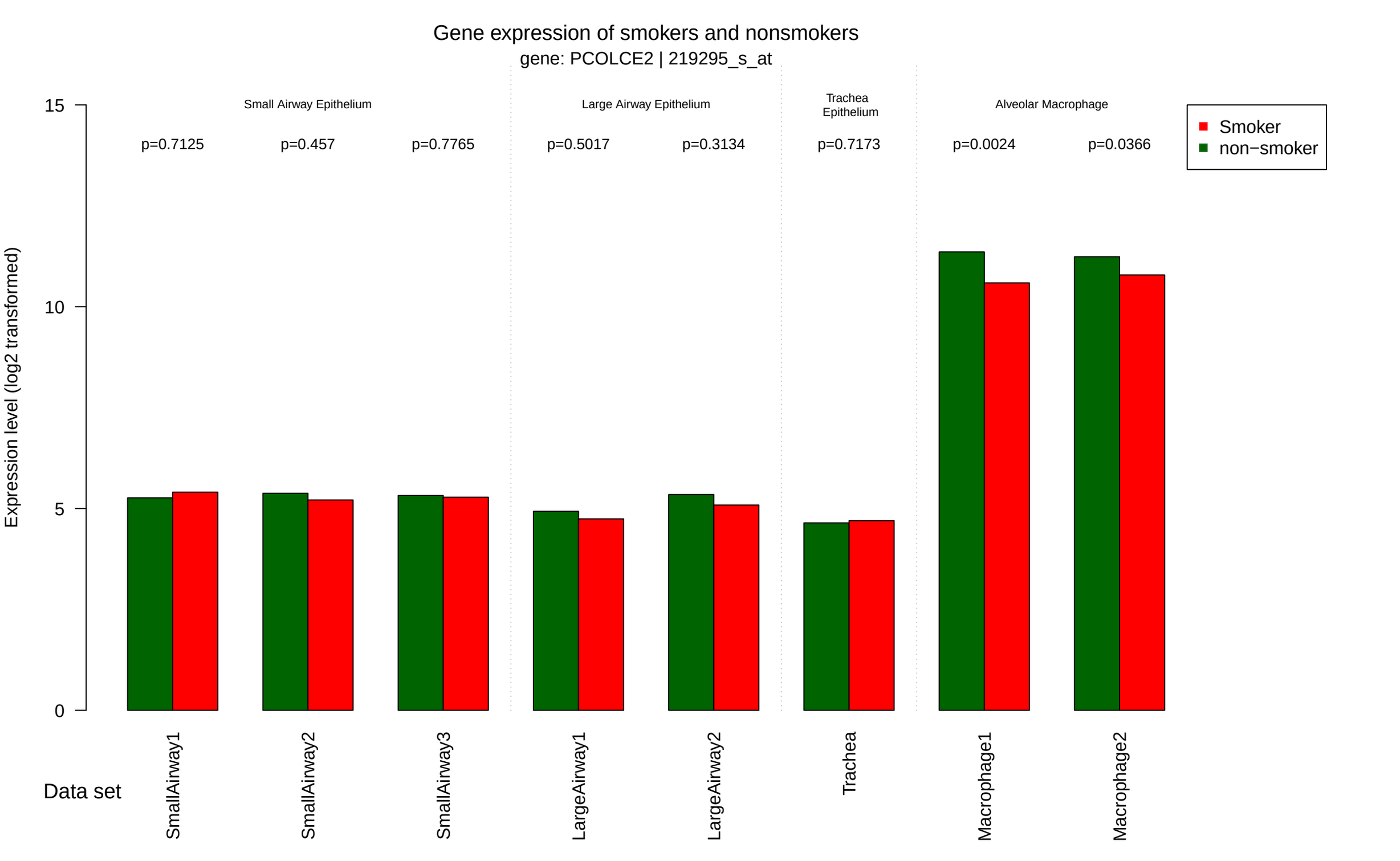
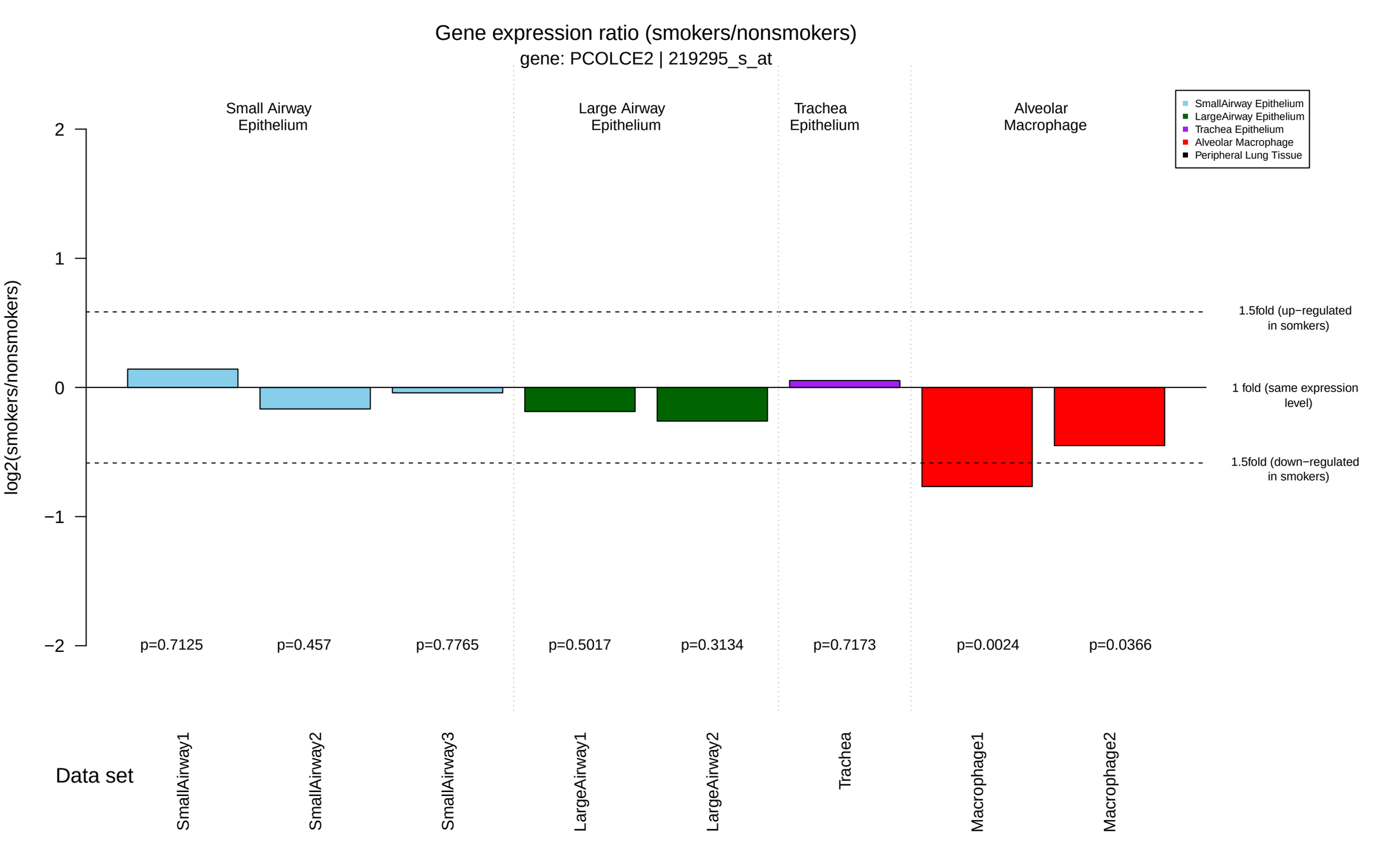
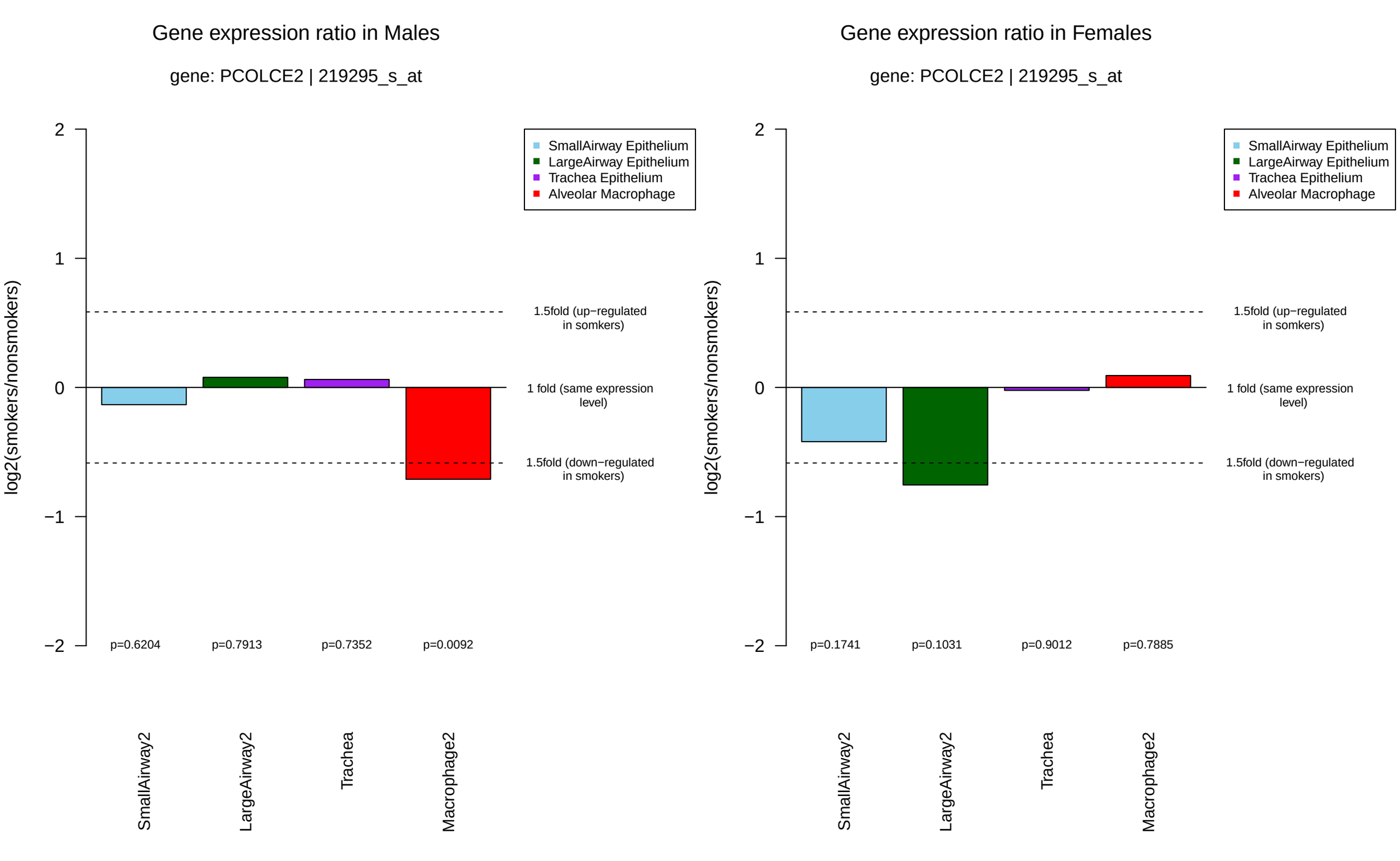
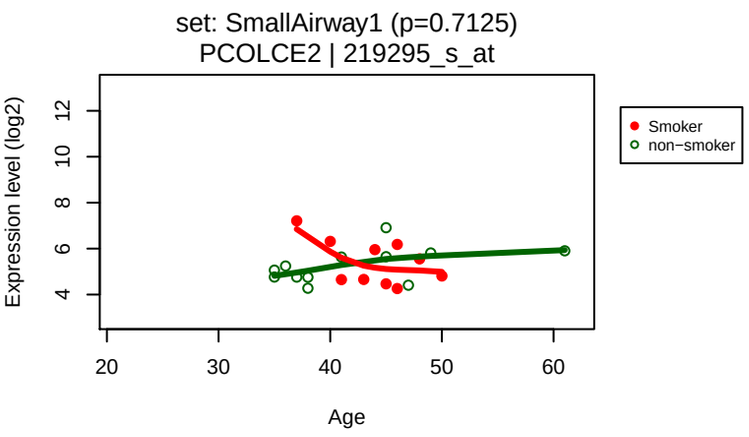
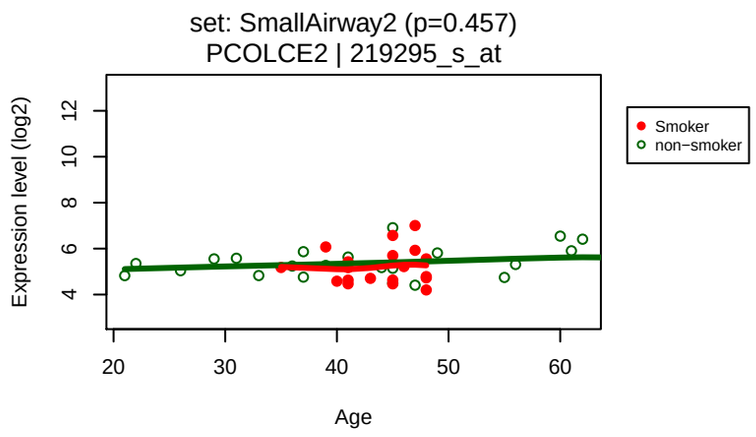
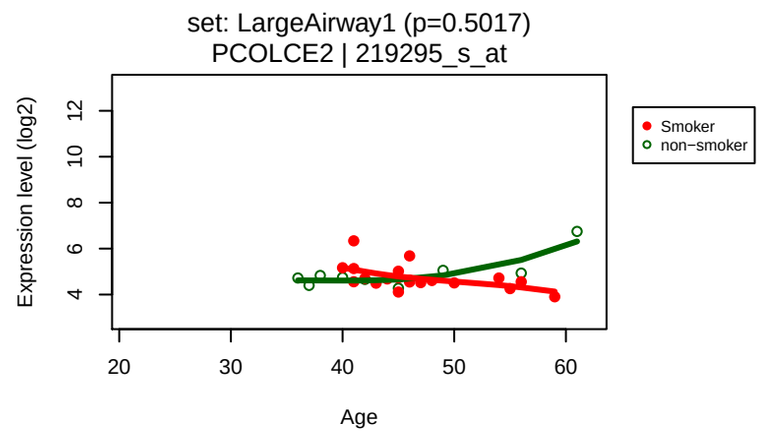

Gene expression of smokers and nonsmokers
gene: PCOLCE2 | 219295_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7125
p=0.457
p=0.7765
p=0.5017
p=0.3134
p=0.7173
p=0.0024
p=0.0366
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

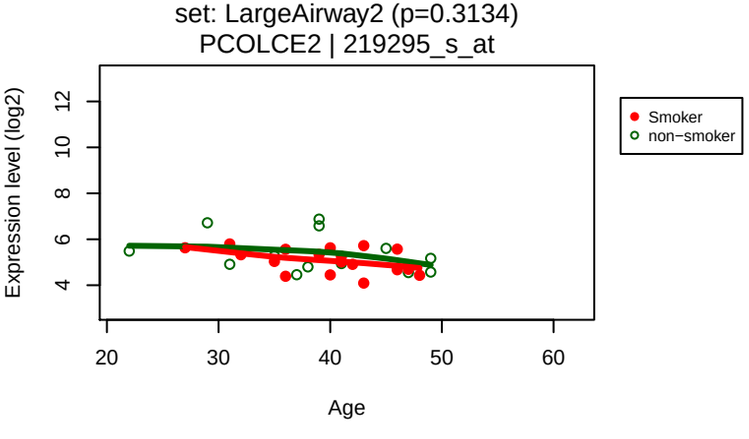

Gene expression ratio (smokers/nonsmokers)
gene: PCOLCE2 | 219295_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7125
p=0.457
p=0.7765
p=0.5017
p=0.3134
p=0.7173
p=0.0024
p=0.0366
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

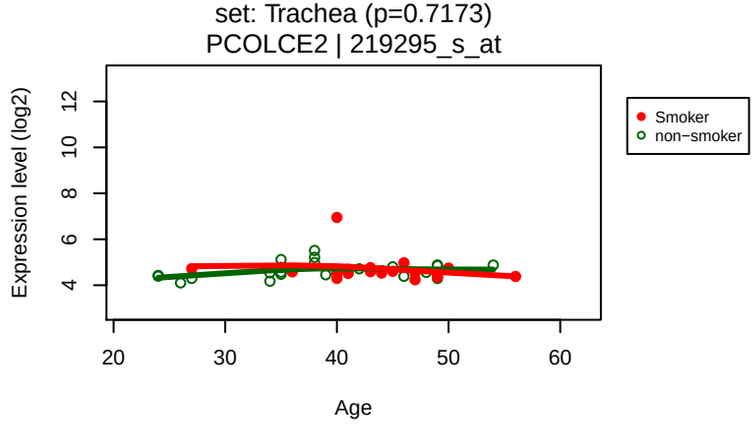

Gene expression ratio in Males
gene: PCOLCE2 | 219295_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6204
p=0.7913
p=0.7352
p=0.0092
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PCOLCE2 | 219295_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1741
p=0.1031
p=0.9012
p=0.7885
SmallAirway2
LargeAirway2
Trachea
Macrophage2

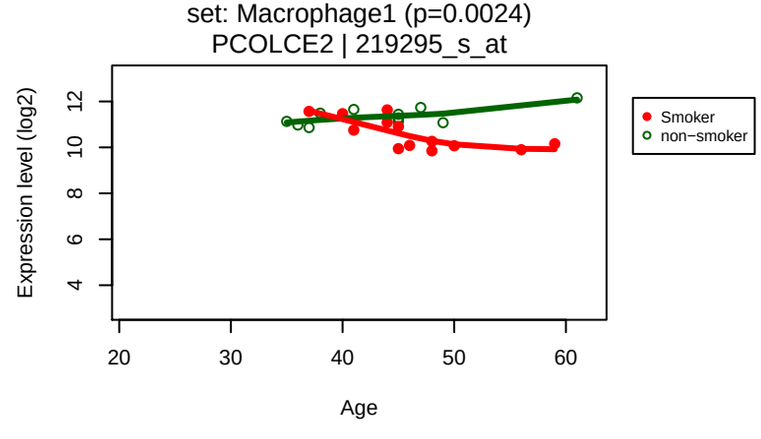

set: SmallAirway1 (p=0.7125)
PCOLCE2 | 219295_s_at
Expression level (log2)
Age
Smoker
non−smoker

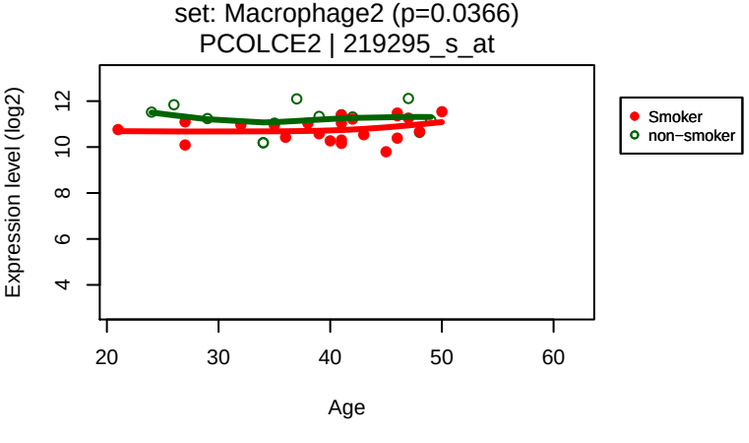

set: SmallAirway2 (p=0.457)
PCOLCE2 | 219295_s_at
Expression level (log2)
Age
Smoker
non−smoker

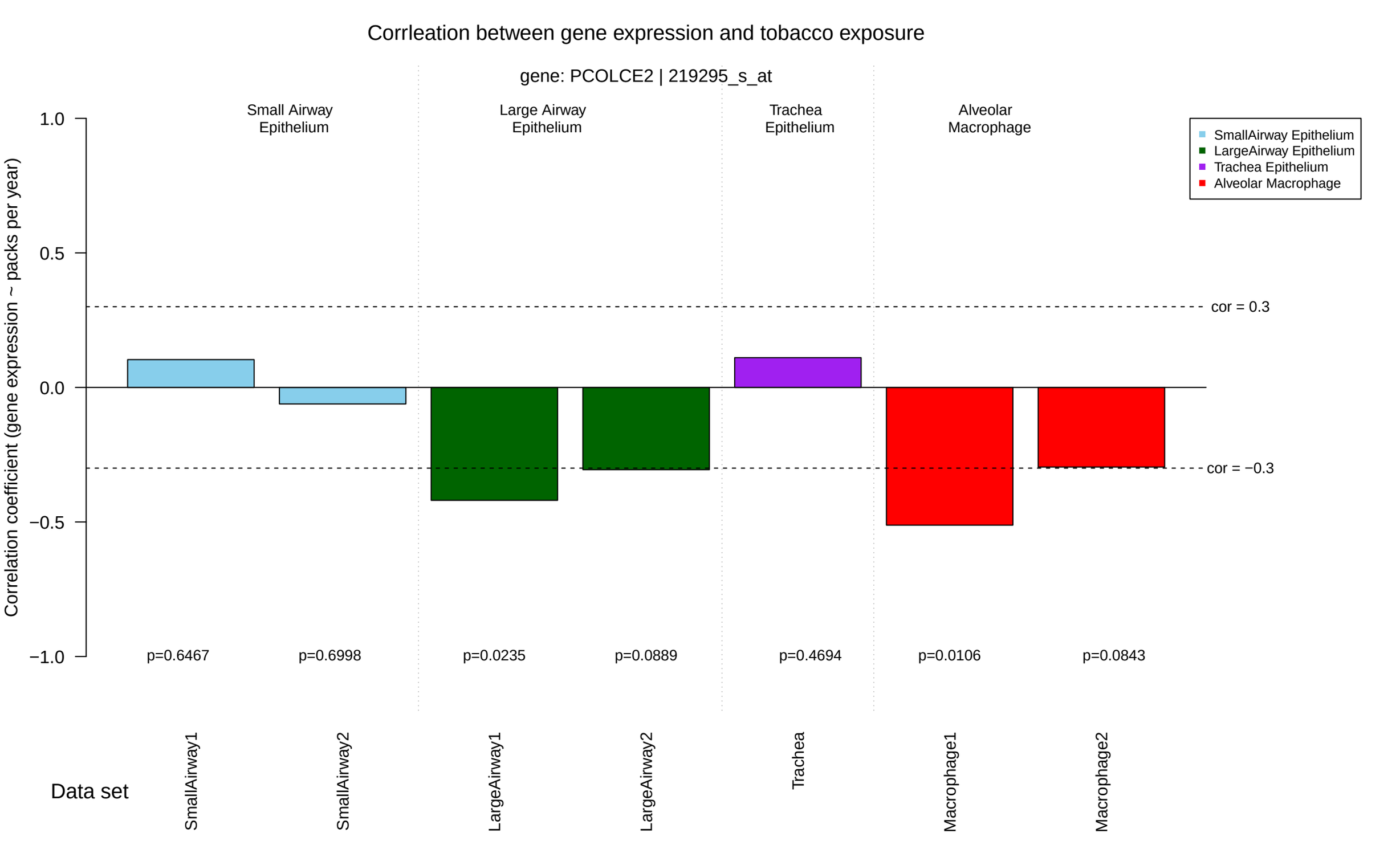

set: LargeAirway1 (p=0.5017)
PCOLCE2 | 219295_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3134)
PCOLCE2 | 219295_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7173)
PCOLCE2 | 219295_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0024)
PCOLCE2 | 219295_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0366)
PCOLCE2 | 219295_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PCOLCE2 | 219295_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6467
p=0.6998
p=0.0235
p=0.0889
p=0.4694
p=0.0106
p=0.0843
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2